



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 28th July, 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : SS G562
Course Title : SOFTWARE ENGINEERING AND MANAGEMENT
Instructor-in-Charge : Dr. Tanmaya Mahapatra (Email: tanmaya.mahapatra@pilani.bits-pilani.ac.in)

1. Course Description:

Software engineering is a discipline that allows us to apply engineering and computer science concepts in the development and maintenance of reliable, usable, and dependable software. There are several areas to focus on within software engineering, such as requirement specification, design, development, testing, maintenance and project management. Software development outside of the classroom is a very complex process, mostly because real-world software is much larger and more complex. The purpose of this course is to present software engineering as a body of knowledge. The course is designed to present software engineering concepts and principles in parallel with the software development life cycle.

2. Scope & Objective of the Course:

This course provides an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics. This also develops an ability to function effectively in a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives. In addition, this course also facilitates an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

3. Learning Outcomes

Upon successful completion of this course, students will be able to:

- Understand and apply fundamental software engineering principles throughout the software development life cycle.
- Identify, formulate, and solve complex software engineering problems using appropriate engineering, scientific, and mathematical principles.
- Elicit, specify, and analyze software requirements for real-world systems.
- Design structured, modular, and maintainable software systems.
- Develop, test, and maintain software using modern tools and best practices.

- Function effectively as a member or leader of a diverse team, demonstrating collaboration and project management skills.
- Plan and manage software projects, including task estimation, scheduling, and risk analysis.
- Demonstrate awareness of ethical, legal, and societal issues in software engineering.

4. Textbooks:

TB1. *Software Engineering*, Sommerville I., 10th, Pearson Education, 2010.

TB2. *Software Engineering: A Practitioner's Approach*, Pressman, R.S., McGraw Hill International Edition, 2010.

References:

Ref1. *An Integrated Approach to Software Engineering*, Jalote, P., 2nd Edition, Narosa, 1998.

Ref2. *Software Engineering*, Schach, S., Tata MCgraw Hill, 7th Ed., 2007.

Ref3. *Software Engineering Fundamentals*, Behforooz, A. & F.J. Hudson, OUP, 2006.

Ref4. *Foundations of Software Engineering*, Ahmed A, CRC Press, 2016.

Additional Readings as assigned and/or made available online by the I/C.

5. Course Modules:

Module #	Module	Learning Objective and Outcomes
1	Background and Introduction (2 Lectures)	To understand the basic concepts and motivation of this course.
2	Software Development Life-cycle Models (3 Lectures)	To understand numerous lifecycle models utilized for maintaining and developing software. To understand how to select an appropriate life-cycle model for a particular project.
3	Requirements Analysis and Specifications (6 Lectures)	To understand the customer requirements and to systematically organize the requirements into a document using Software Requirement Specification and Formal Notations.
4	Software Project Management (10 Lectures)	To understand how to enable a group of developers to work effectively towards the successful completion of a project.

5	Software Design (5 Lectures)	To understand the activities carried out during the design phase and how to transform the generated specifications into the design document. To understand how to decompose a problem into different modules and how to identify the relationships and the associated interfaces among several modules.
6	Software Reliability (2 Lectures)	To understand how to quantitatively measure the reliability of a software product and how to predict when a given level of reliability will be achieved.
7	Software Coding and Testing (14 Lectures)	To understand how to transform the design of a system into some code in high-level language and then to unit test this code. To understand how to detect bugs in a program using different testing techniques.

6. Course Plan:

Lecture No.	Modules	Topics
1–2	Background, Motivation and Introduction	Evolution of an Art into an Engineering Discipline, What is Software Engineering, Exploratory Style of Software Development, Principles of Software Engineering to Overcome Human Cognitive Limitations, Rise of Software Engineering
3–5	Software Lifecycle Models	Introduction, Waterfall Model and its Variants: Classical Waterfall Model, Iterative Waterfall Model, V-Model, Prototyping Model, Incremental Development Model and Evolutionary Model. Rapid Application Development (RAD) Model, Spiral Model, Agile Development Models, Selection of Appropriate Lifecycle model for a project
6–11	Requirement Analysis and Specifications	Requirement Gathering and Analysis, Software Requirement Specification (SRS), Techniques for Handling Complex Logic through Semi-Formal techniques: Decision Trees and Tables, Formal System Specifications using Algebraic Specifications and Axiomatic Specifications.
12–21	Software Project Management Techniques	Complexities of Software Project Management, Techniques for Project Size Estimation: Lines of Code (LOC) and Function Point (FP).

Lecture No.	Modules	Topics
		Project Estimation Techniques: Empirical Estimation Techniques, Heuristic Estimation Techniques: COCOMO, Complete COCOMO and COCOMO 2, Analytical Technique: Halstead's Software Science. Staffing level Estimation, Project Scheduling: Work Break-down Structure (WBS), Activity Networks, Critical Path Method, PERT Chart, Earned Value Analysis, Project Crashing
22–26	Software Design	Outcome of Design Process, Classification of Design Methodologies, Characteristics of a Good Design, Modularity of Design, Cohesion and Coupling, Classification of Cohesion and Coupling, Function-Oriented Design and Object-Oriented Design
27–28	Software Reliability	Software Reliability: Hardware versus Software Reliability, Reliability Metrics, Reliability Growth Models: Jelinski and Moranda Model, Littlewood and Verall's Model, Basic Execution Model and Logarithmic Poisson Execution Model
29–43	Software Coding and Testing	Software Coding: Errors, Faults and Failures, Techniques for Reducing the Bugs in Program. Software Testing: Verification and Validation, Pesticide Effect in Testing, Test Cases, Minimal Test Suite, Unit Testing, Black-box Testing Strategies: Equivalence Class Partitioning Testing and its Derivatives.

7. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Software Development Project (team)	30	take home	Abstract: 25.08.2026; Final: 10.11.2026	Code & Documentation to be submitted, followed by presentation and Viva
Research Seminar (individual)	10	take home	10.11.2026	take-home; Paper submission followed by presentation & viva. Details to be announced.
Mid-Semester Examination	25	90 mins.	Refer to Time table	Partly Open Book; All formulas in printed sheets allowed + Calculator
Comprehensive Examination	35	180 mins.	Refer to Time table	Partly Open Book; All formulas in printed sheets allowed + Calculator

Software Development Project (team-based) A complete project is to be done by a team of students in two Stages using the practices of Software Engineering covered in the regular lectures. In Stage I, the project will aim to provide you hands-on experience with different aspects of Software Engineering including designing of informal specifications, semi-formal specifications and formal specifications using SRS document, decision table and algebraic specifications. Stage II include designing of DFD modelling, Activity Network, Work Breakdown Structure and PERT charts. In addition, you will also have to design the test cases corresponding to the working prototype of your project. Overall, you will be working on these aspects of an assigned project to get a feel of the real-life software development process. Evaluation will be done continuously, on the basis of the quality of work products delivered according to the project plan and schedule, as well as process compliance. Grades assigned to individual students are determined using periodic presentations, design and other documents, teamwork, quality of the prototype and the product, and technological innovation.

Research Seminar (individual) A pool of research papers will be provided to the students, from which each student must select one paper on a first-come, first-served basis. The selected paper should be studied in depth to understand its research problem, methodology, and contributions. Subsequently, the student is required to identify and collect additional relevant research articles on the same topic by applying a snowballing approach through cited and citing references. Based on this curated set of papers, the student must critically analyse and synthesize the literature to prepare a comprehensive research summary. The final deliverable is a structured academic report of 4–6 pages, excluding references, prepared using the approved LaTeX template. The report should demonstrate clarity of understanding, coherence, and critical insight, and must be submitted along with a plagiarism report indicating similarity below 20%. This activity is designed to strengthen students' research literacy, analytical thinking, and technical writing skills while fostering engagement with contemporary scholarly work relevant to the course domain. The component ends with a final presentation and viva.

NC Criterion: As per BITS academic regulations

8. Office Consultation Hour:

Tanmaya Mahapatra **Monday 11 AM to 12 noon – Chamber: 6121 - Y**

Please email for an appointment to meet at other timings.

9. Make-up Policy:

- Mid-Sem Exam has one make-up. This means, that a student can abstain herself/himself (on valid reasons) in this Test. This make up would be conducted immediately after regular mid-sem exam and would include complete syllabus. Of course, the student would apply for the make up in advance with documentary proof.
- Only on producing documentary proof of possible absence, which proves that student would be physically unable to appear for the test/exam, the decision of granting the make-up will be taken.
- Prior Permission of AGSRD is required to get make-up for the comprehensive exam.

Tanmaya Mahapatra
Instructor-in-charge
SS G562 – SEM